

Name: _____

Score: _____ / _____

Quiz 4

Part 1: Quiz 4

1

The class P can contain languages that are not decidable in polynomial time by a Non-Deterministic Turing machine.

True

False

Answer Point Value: 1.0 points

Answer Key: False

2

The Halting Problem is NP-Hard.

A. True

B. False

C. Not Known

Answer Point Value: 1.0 points

Answer Key: A

3

We can find the prime factors of an integer in polynomial time using a Deterministic Turing Machine.

A. True

B. False

C. Not Known

Answer Point Value: 1.0 points

Answer Key: C

4

The Strong Church Turing Thesis is known to be False.

True

False

Answer Point Value: 1.0 points

Answer Key: False

5

NP-Complete problems can be decided in polynomial time on a Deterministic Turing Machine.

A. True

B. False

C. Not Known

D. None of the above

Answer Point Value: 1.0 points

Answer Key: C

6

NP-Hard problem can always be decided in finite time.

A. True

B. False

C. Not Known

D. None of the above

Answer Point Value: 1.0 points

Answer Key: B

7

Proving that the Factorization problem is in the class P implies $P = NP$.

A. True

B. False

C. Not Known

D. None of the above

Answer Point Value: 1.0 points

Answer Key: C

8

An exponential lower bound on an NP-Complete problem implies that P is not equal to NP.

- A. True
- B. False
- C. Not Known
- D. None of the Above

Answer Point Value: 1.0 points

Answer Key: A

9

$(L = \{x \mid x \text{ ends with } 0 \text{ and } (P = NP), \text{ or } (x \text{ ends with } 1 \text{ and } (P \neq NP))\})$ is decidable.

- A. True
- B. False
- C. Not Known

Answer Point Value: 1.0 points

Answer Key: A

10

The set of statements that are provable in ZF set theory is Turing recognizable.

True

False

Answer Point Value: 1.0 points

Answer Key: True

11

There exists an (NP) language that's neither in (P) nor (NPC) .

- A. True
- B. False
- C. Not Known

Answer Point Value: 1.0 points

Answer Key: C

Every consistent axiomatic theory is sound.

A. True

B. False

C. Not Known

Answer Point Value: 1.0 points

Answer Key: B